

Econometrics Project - R-studio analysis process

2024-12-03

Package Loading

```
library(tidyverse)
library(modelsummary)
library(fixest)
library(ggplot2)
library(readxl)
library(GGally)
library(dplyr)
```

Data Preparation

```
# Import dataset and rename columns
data <- read_excel("2019_data.xlsx", sheet = "Data")

data <- data %>%
  rename(
    country = `Country Name`,
    fdi = `Foreign direct investment, net inflows (current US$)`,
    eodb = `Ease of doing business score (0 = lowest performance to 100 = best performance)`,
    gdp = `GDP per capita`,
    unemployment = `Unemployment, total (% of total labor force)`,
    urban = `Urban population (% of total population)`,
    political = `Political Stability and Absence of Violence/Terrorism`,
    corruption = `Control of Corruption`
  )
```

What happens in the code:

The data is loaded from an Excel file, specifically from the sheet named **“Data”**, and stored in the object data.

Renaming columns:

Columns are renamed for easier analysis, such as:

- country — the name of the country,
- fdi — foreign direct investment (current US\$)
- etc.

Descriptive Statistics

- Purpose: Summarize key indicators (e.g., FDI, Ease of Doing Business, GDP).
- A statistical overview includes count, mean, median, standard deviation, minimum, and maximum values of variables.

Data Summary

```
datasummary(fdi + eodb + gdp + unemployment + urban + political + corruption ~ N
             + Mean + Median + SD + Min + Max,
             data = data)
```

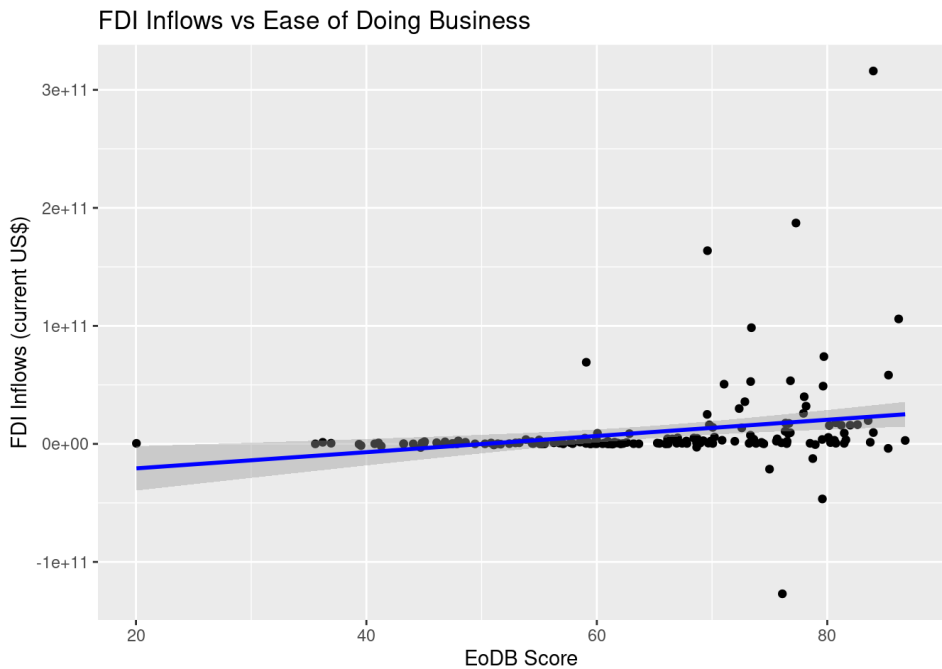
	N	Mean	Median		SD	Min	Max
fdi	171	9712166106.95	1024779237.91	35971385433.36 (tel:35971385433.36)	-127018529907.98	3.2e+11	
eodb	171	64.37	65.97		13.04	20.04	86.76
gdp	171	14625.67	5807.84		20056.15	216.97	112726.44
unemployment	171	7.16	5.02		5.63	0.10	29.05
urban	171	59.21	60.04		22.78	13.25	100.00
political	171	-0.12	-0.08		0.90	-2.65	1.62

Graphs

Scatterplots and histograms visualize relationships between variables like Ease of Doing Business (EoDB) and FDI inflows.

```

# Scatterplot: EoDB vs FDI
ggplot(data, aes(x = eodb, y = fdi)) +
  geom_point() +
  geom_smooth(method = "lm", color = "blue") +
  labs(title = "FDI Inflows vs Ease of Doing Business", x = "EoDB Score", y = "FDI Inflows (current US$)")
  
```



```

# Remove rows for Luxembourg and Cyprus
data_clean <- data %>%
  filter(!country %in% c("Luxembourg", "Cyprus", "Hungary"))

# Check if the rows were successfully removed
table(data_clean$country %in% c("Luxembourg", "Cyprus", "Hungary")) # Should return all FALSE
  
```

```

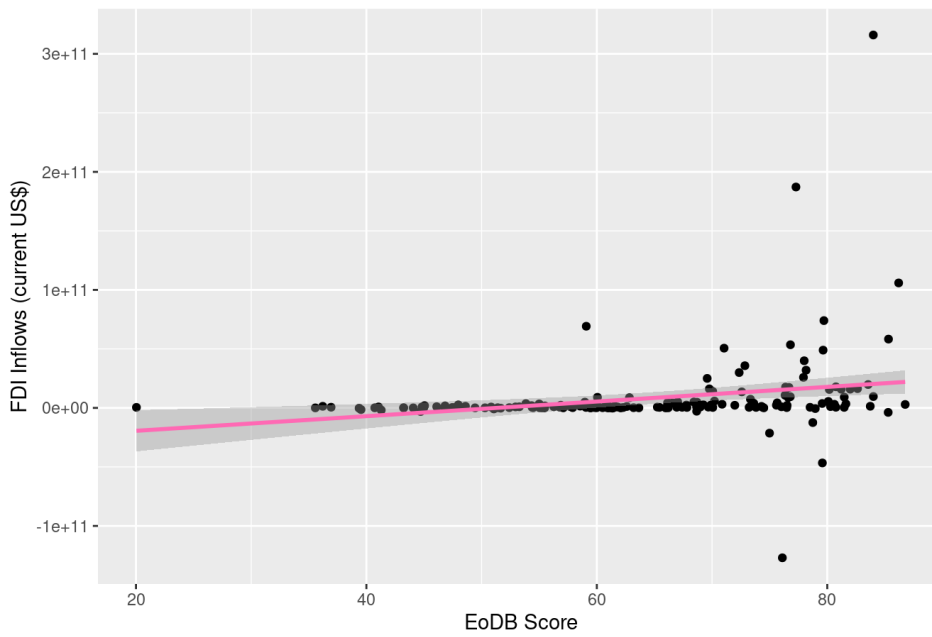
##
## FALSE
## 168
  
```

Outliers are removed (Luxembourg, Cyprus, and Hungary) because their extreme FDI values could distort regression analysis. For example, Luxembourg has exceptionally high FDI due to tax-related anomalies.

```

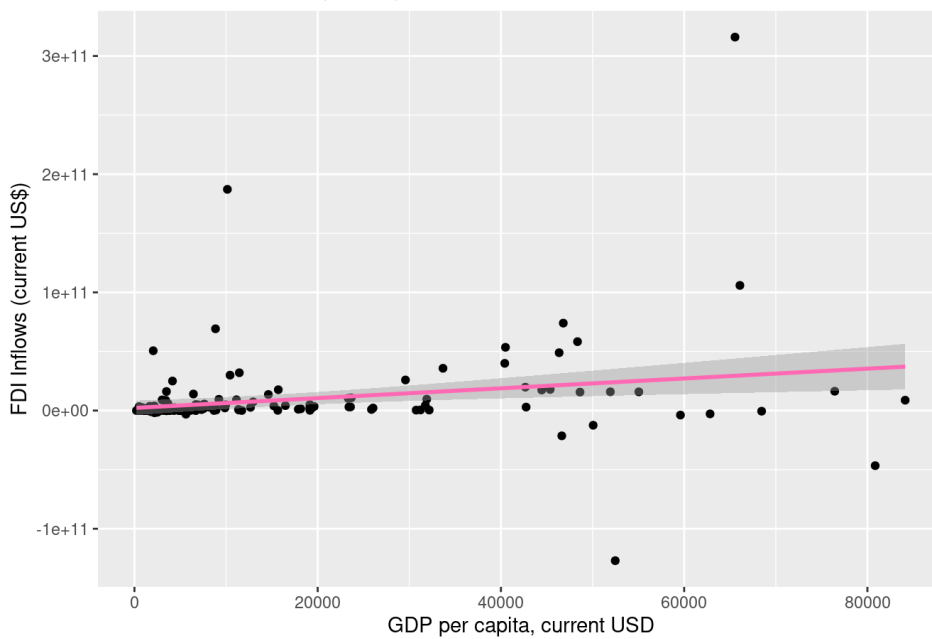
# Scatterplots =====
#EoDB vs FDI
ggplot(data_clean, aes(x = eodb, y = fdi)) +
  geom_point() +
  geom_smooth(method = "lm", color = "hotpink") +
  labs(title = "FDI Inflows vs Ease of Doing Business", x = "EoDB Score", y = "FDI Inflows (current US$)")
  
```

FDI Inflows vs Ease of Doing Business



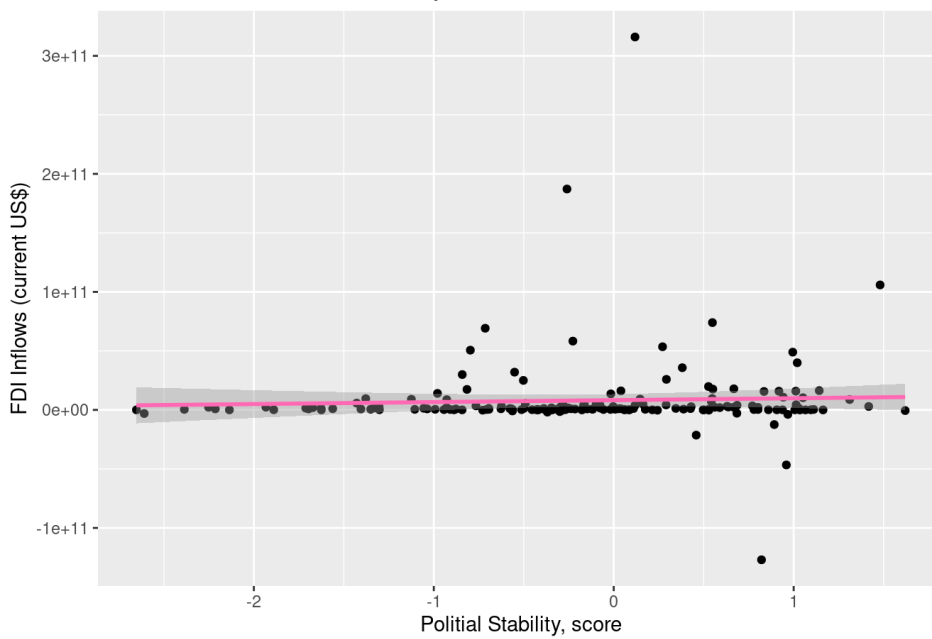
```
#GDP vs FDI
ggplot(data_clean, aes(x = gdp, y = fdi)) +
  geom_point() +
  geom_smooth(method = "lm", color = "hotpink") +
  labs(title = "FDI Inflows vs GDP per capita", x = "GDP per capita, current USD", y = "FDI Inflows (current US
$)")
```

FDI Inflows vs GDP per capita



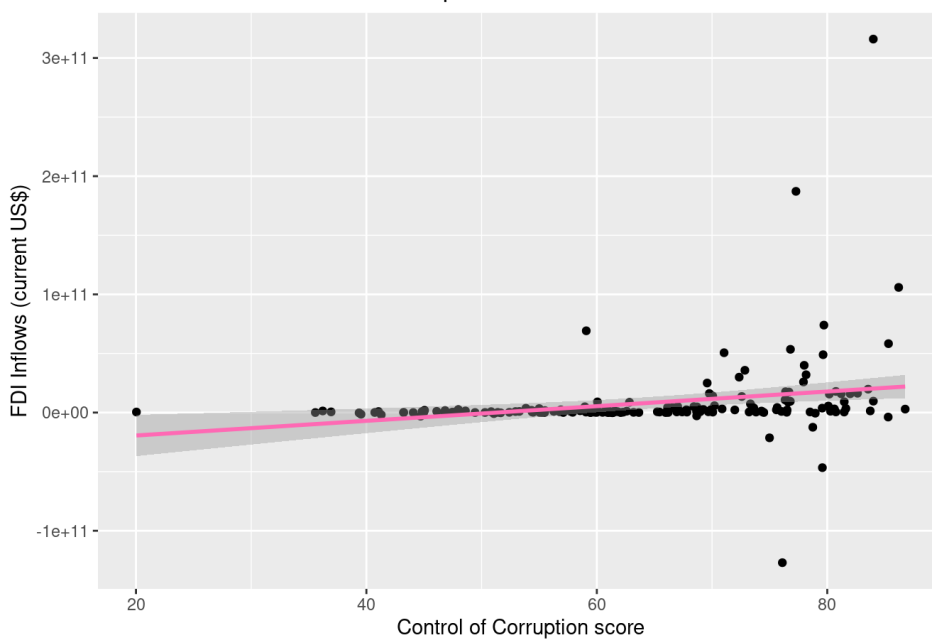
```
#Political Stability vs FDI
ggplot(data_clean, aes(x = political, y = fdi)) +
  geom_point() +
  geom_smooth(method = "lm", color = "hotpink") +
  labs(title = "FDI Inflows vs Political Stability", x = "Politial Stability, score", y = "FDI Inflows (current U
S$)")
```

FDI Inflows vs Political Stability



```
#Control of Corruption vs FDI
ggplot(data_clean, aes(x = eodb, y = fdi)) +
  geom_point() +
  geom_smooth(method = "lm", color = "hotpink") +
  labs(title = "FDI Inflows vs Control of Corruption", x = "Control of Corruption score", y = "FDI Inflows (current US$)")
```

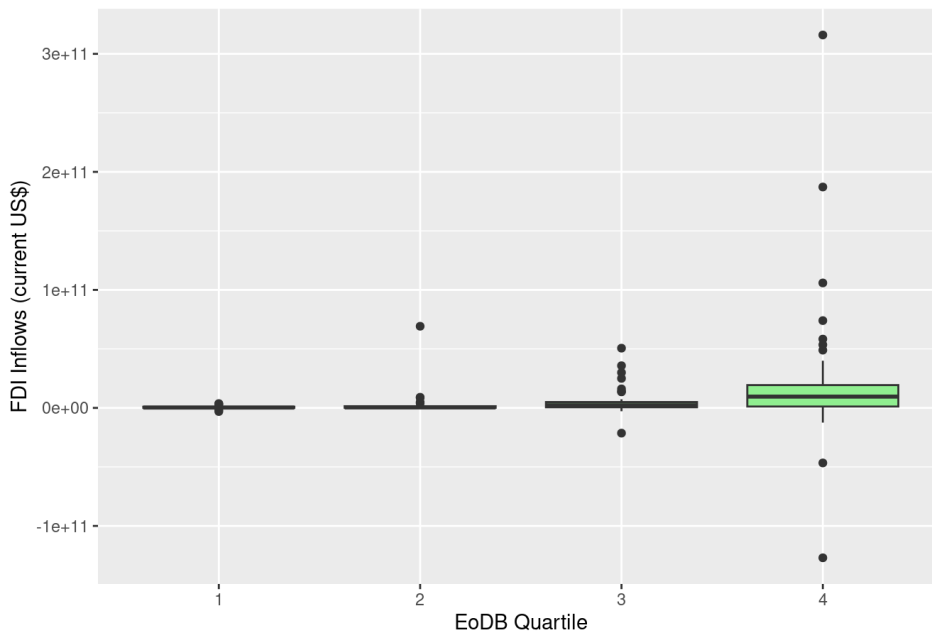
FDI Inflows vs Control of Corruption



```
# Boxplots =====
#FDI inflows by quartile of
data_quartile <- data_clean %>%
  mutate(eodb_quartile = ntile(eodb, 4),
         gdp_quartile = ntile(gdp, 4),
         corruption_quartile = ntile(corruption, 4),
         political_quartile = ntile(political, 4))

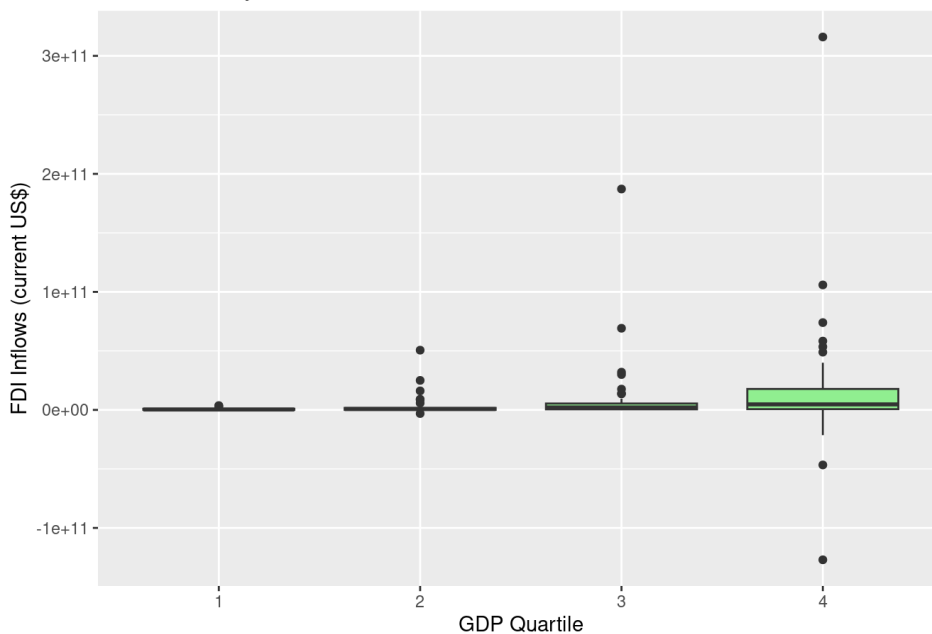
ggplot(data_quartile, aes(x = factor(eodb_quartile), y = fdi)) +
  geom_boxplot(fill = "lightgreen") +
  labs(title = "FDI Inflows by EoDB Quartile", x = "EoDB Quartile", y = "FDI Inflows (current US$)")
```

FDI Inflows by EoDB Quartile



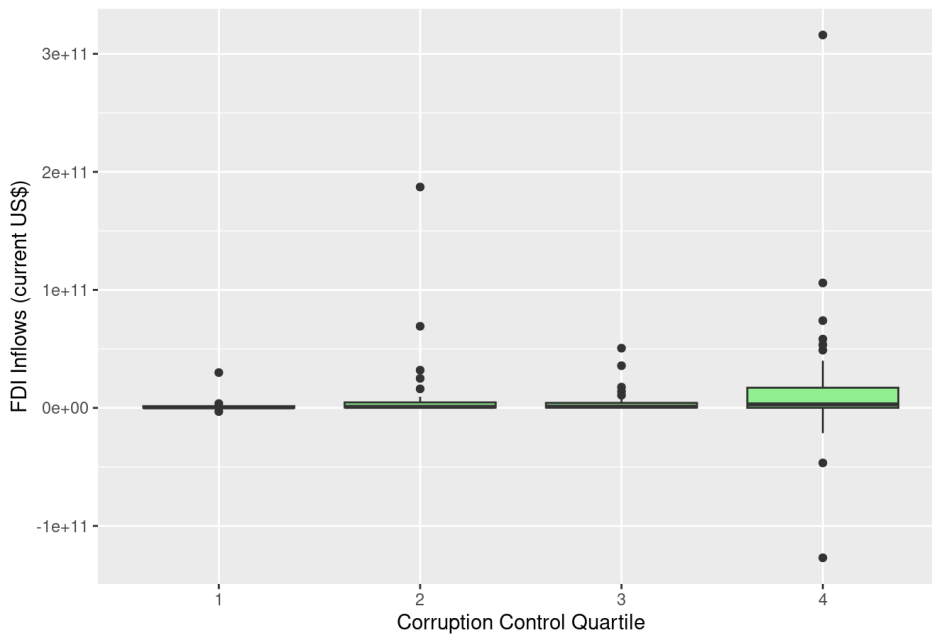
```
ggplot(data_quartile, aes(x = factor(gdp_quartile), y = fdi)) +
  geom_boxplot(fill = "lightgreen") +
  labs(title = "FDI Inflows by GDP Quartile", x = "GDP Quartile", y = "FDI Inflows (current US$)")
```

FDI Inflows by GDP Quartile



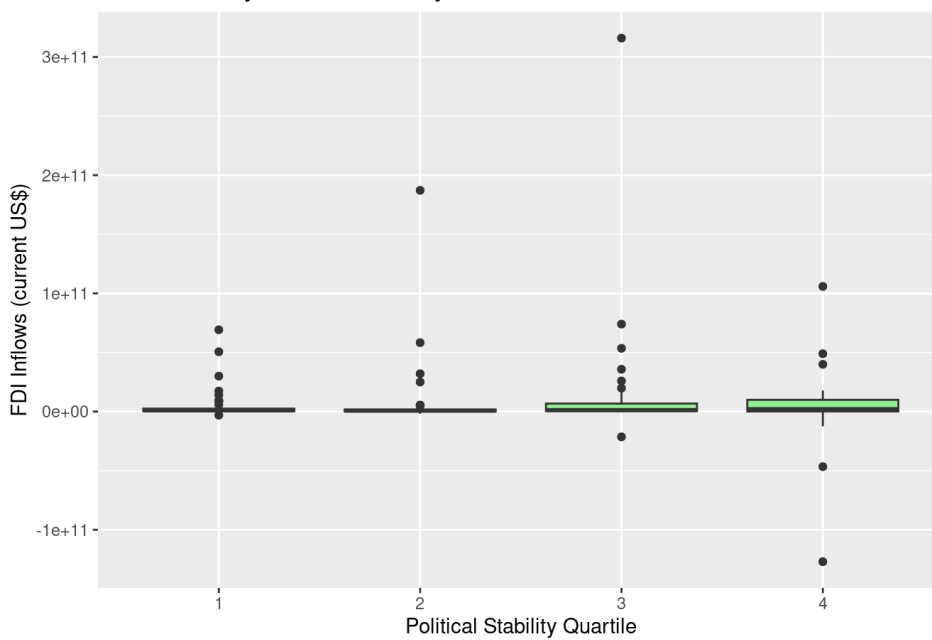
```
ggplot(data_quartile, aes(x = factor(corruption_quartile), y = fdi)) +
  geom_boxplot(fill = "lightgreen") +
  labs(title = "FDI Inflows by Corruption Control Quartile", x = "Corruption Control Quartile", y = "FDI Inflows (current US$)")
```

FDI Inflows by Corruption Control Quartile



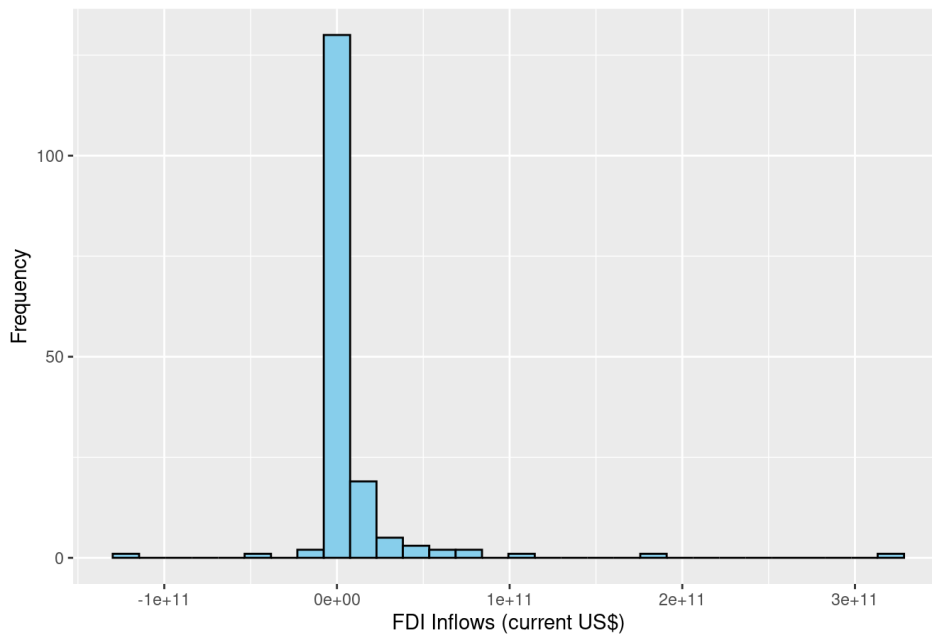
```
ggplot(data_quartile, aes(x = factor(political_quartile), y = fdi)) +
  geom_boxplot(fill = "lightgreen") +
  labs(title = "FDI Inflows by Political Stability Quartile", x = "Political Stability Quartile", y = "FDI Inflows (current US$)")
```

FDI Inflows by Political Stability Quartile



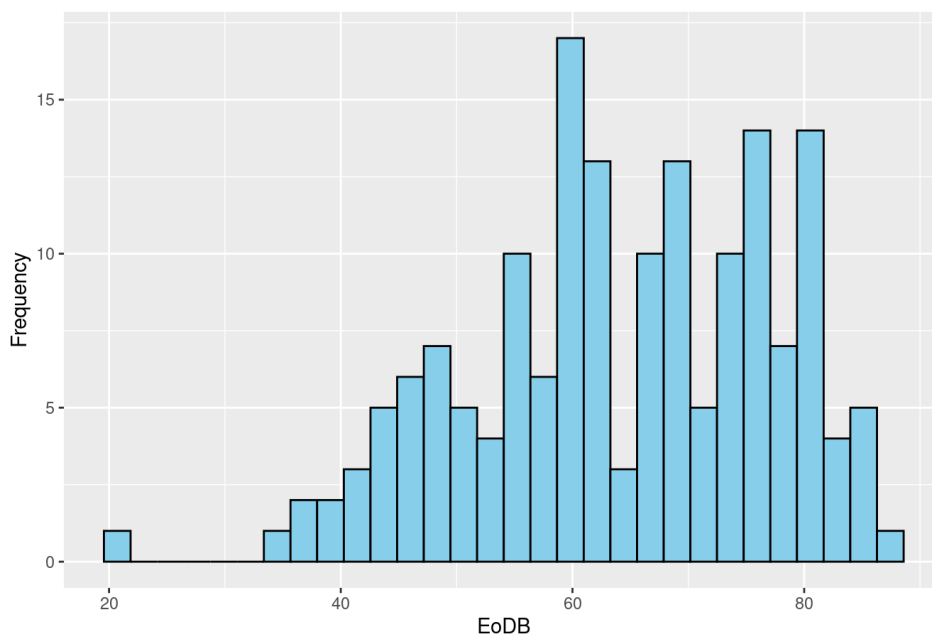
```
# Histograms =====
# Histogram: FDI inflows
ggplot(data_clean, aes(x = fdi)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 30) +
  labs(title = "Distribution of FDI Inflows", x = "FDI Inflows (current US$)", y = "Frequency")
```

Distribution of FDI Inflows



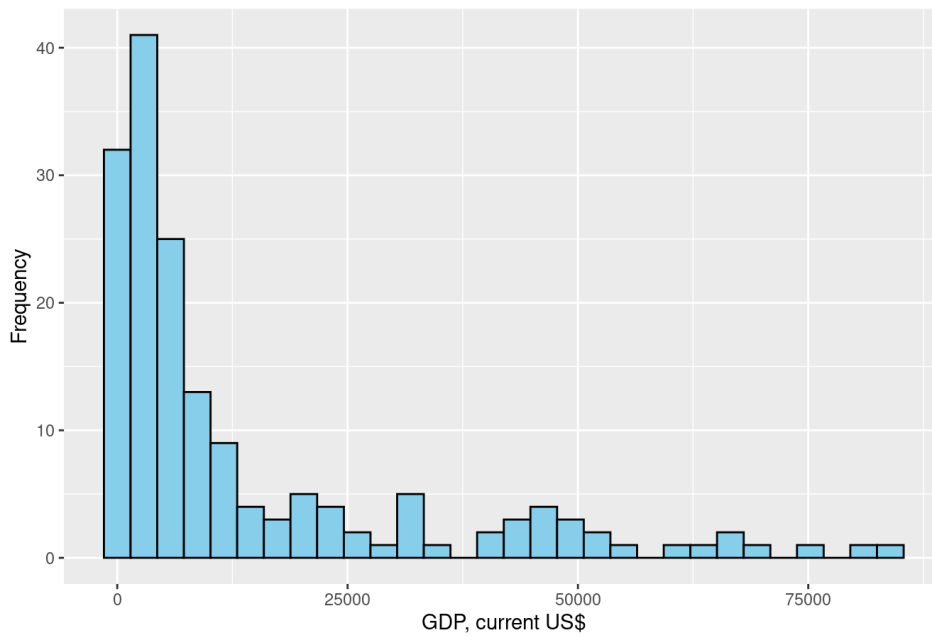
```
# Histogram: Ease of Doing Business
ggplot(data_clean, aes(x = eodb)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 30) +
  labs(title = "Distribution of EoDB", x = "EoDB", y = "Frequency")
```

Distribution of EoDB



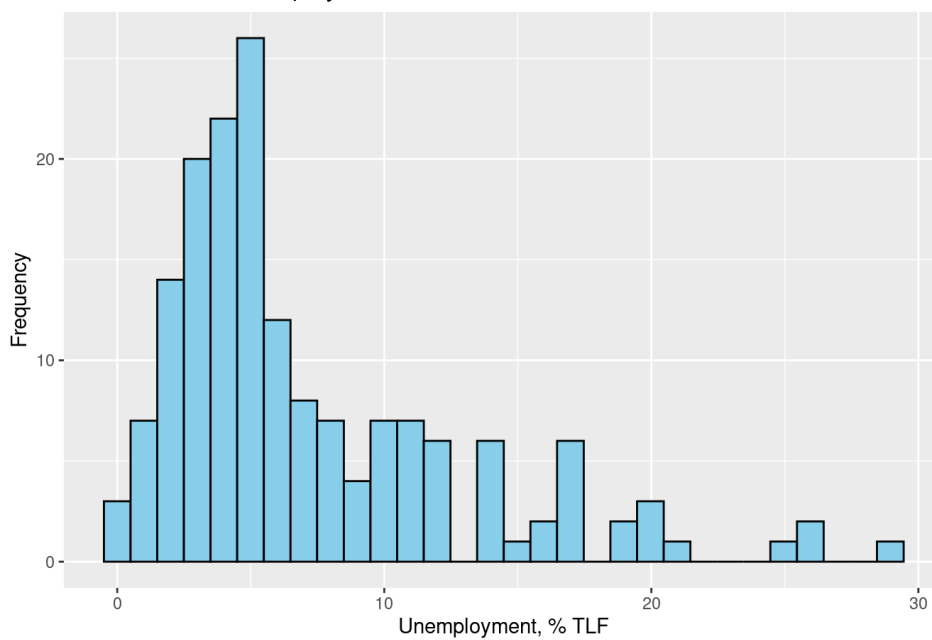
```
# Histogram: GDP per capita
ggplot(data_clean, aes(x = gdp)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 30) +
  labs(title = "Distribution of GDP per capita", x = "GDP, current US$", y = "Frequency")
```

Distribution of GDP per capita

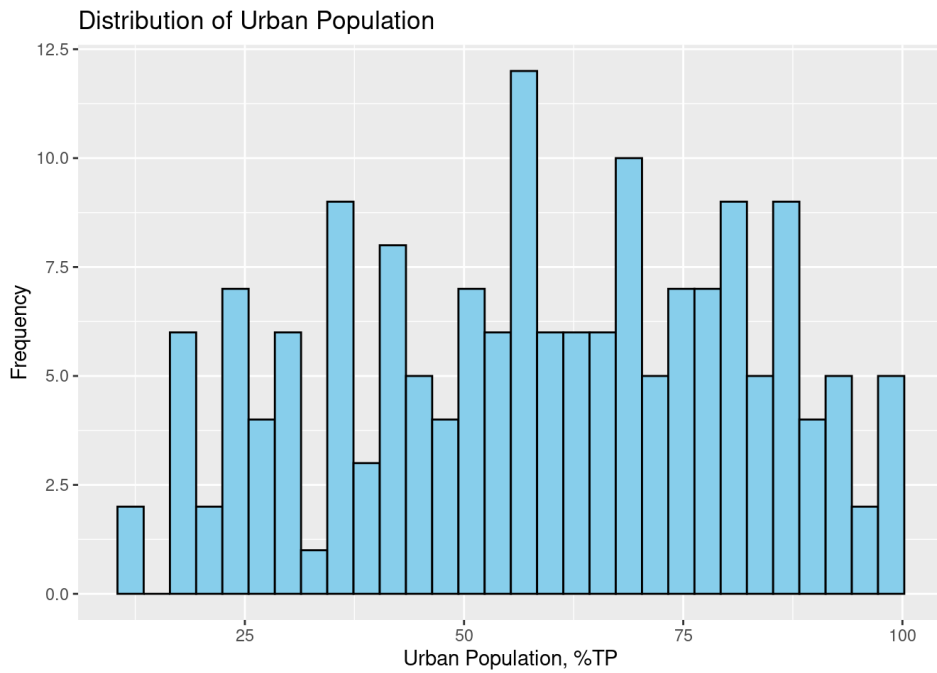


```
# Histogram: Unemployment
ggplot(data_clean, aes(x = unemployment)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 30) +
  labs(title = "Distribution of Unemployment", x = "Unemployment, % TLF", y = "Frequency")
```

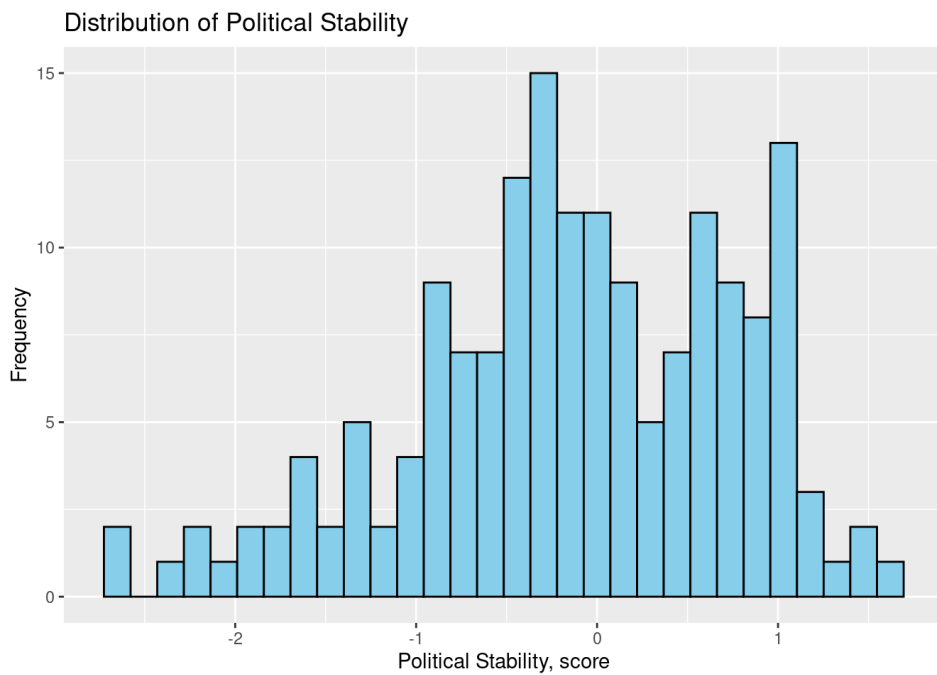
Distribution of Unemployment



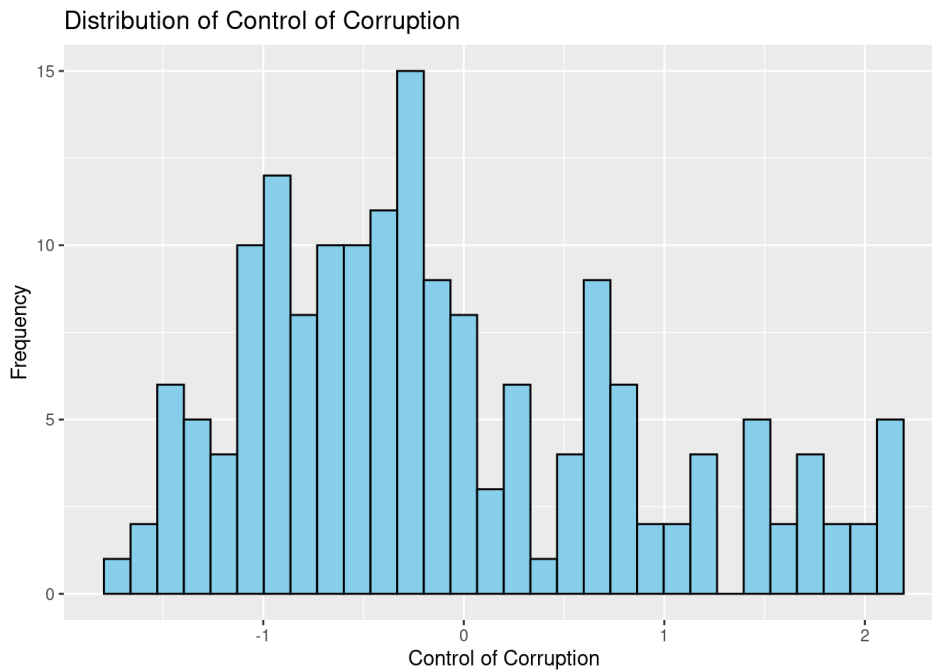
```
# Histogram: Urban Population
ggplot(data_clean, aes(x = urban)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 30) +
  labs(title = "Distribution of Urban Population", x = "Urban Population, %TP", y = "Frequency")
```

```
# Histogram: Political Stability
ggplot(data_clean, aes(x = political)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 30) +
  labs(title = "Distribution of Political Stability", x = "Political Stability, score", y = "Frequency")
```



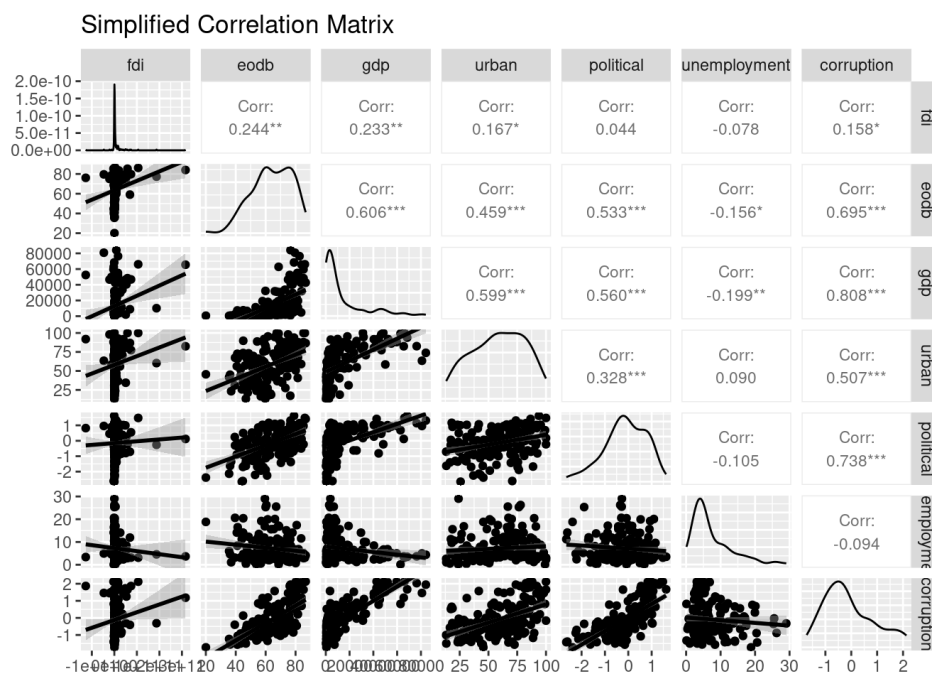
```
# Histogram: Control of Corruption
ggplot(data_clean, aes(x = corruption)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 30) +
  labs(title = "Distribution of Control of Corruption", x = "Control of Corruption", y = "Frequency")
```



Correlation Matrix

- Purpose: Examine relationships between variables.
- The `ggpairs` function creates a correlation matrix, providing scatterplots, correlation coefficients, and density plots for variables like FDI, EoDB, GDP, etc.

```
ggpairs(
  data_clean,
  columns = c("fdi", "eodb", "gdp", "urban", "political",
             "unemployment", "corruption"),
  upper = list(continuous = wrap("cor", size = 3)), # Show correlations
  lower = list(continuous = "smooth"),             # Smoothed scatterplots
  diag = list(continuous = "densityDiag")          # Density plots
) +
  ggtitle("Simplified Correlation Matrix")
```



Regression Analysis

Purpose: Explore relationships between FDI and other variables through econometric modeling.

Preparation: Create Log Variables

```
# Create variables:
data <- data %>%
  mutate(log_fdi = log(fdi))

data <- data %>%
  mutate(log_eodb = log(eodb))
```

Descriptive Statistics for updated data

```
datasummary_skim(data)
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
fdi	171	0	9712166106.9	3.597139e+10	-1.270185e+11	1024779237.9	3.159840e+11	
eodb	171	0	64.4	1.300000e+01	2.000000e+01	66.0	8.680000e+01	
gdp	171	0	14625.7	2.005610e+04	2.170000e+02	5807.8	1.127264e+05	
unemployment	169	0	7.2	5.600000e+00	1.000000e-01	5.0	2.900000e+01	
urban	168	0	59.2	2.280000e+01	1.320000e+01	60.0	1.000000e+02	
political	171	0	-0.1	9.000000e-01	-2.700000e+00	-0.1	1.600000e+00	
corruption	171	0	-0.1	1.000000e+00	-1.700000e+00	-0.3	2.100000e+00	
log_fdi	156	9	21.1	2.200000e+00	1.530000e+01	21.0	2.650000e+01	
log_eodb	171	0	4.1	2.000000e-01	3.000000e+00	4.2	4.500000e+00	

Regressions:

```
# Simplified Model
modell <- feols(fdi ~ eodb, data = data, vcov = "hetero")
```

```
modell
```

```
## OLS estimation, Dep. Var.: fdi
## Observations: 171
## Standard-errors: Heteroskedasticity-robust
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -34469765733 13804220064 -2.49705 0.0134805 *
## eodb         686404628   250786970  2.73700 0.0068636 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 3.474e+10 Adj. R2: 0.05635
```

- What we are doing: A simple linear regression where FDI is regressed on EoDB to examine the direct relationship between economic freedom and FDI inflows
- Components: Dependent variable: fdi, Independent variable: eodb. Justification: This serves as a baseline to test the primary hypothesis of the project's focus
- Purpose: To establish if EoDB alone significantly impacts FDI

```
# Baseline Model
modell2 <- feols(fdi ~ eodb + gdp + unemployment +
  urban + political + corruption, data = data, vcov = "hetero")
```

```
modell2
```

```
## OLS estimation, Dep. Var.: fdi
## Observations: 171
## Standard-errors: Heteroskedasticity-robust
##           Estimate   Std. Error   t value Pr(>|t|)
## (Intercept) -36063998392 1.706546e+10 -2.113275 0.036091 *
## eodb         536030990 2.807191e+08  1.909492 0.057945 .
## gdp          806195 5.397212e+05  1.493725 0.137169
## unemployment -4804530 2.713487e+08 -0.017706 0.985895
## urban        -25345998 1.386132e+08 -0.182854 0.855138
## political    -3752885308 3.248936e+09 -1.155112 0.249725
## corruption   -7832559359 6.052766e+09 -1.294046 0.197469
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 3.325e+10   Adj. R2: 0.108883
```

- What we are doing: Adding control variables (gdp, unemployment, urban, political, corruption) to isolate the effect of EoDB on FDI
- Components: Dependent variable: fdi, Independent variables: eodb and controls
- Justification: Control variables capture key economic, social, and institutional factors that influence FDI inflows
- Purpose: To assess the robustness of the relationship between EoDB and FDI

```
# Log-Lin Model
model3 <- feols(log_fdi ~ eodb + gdp + unemployment +
                urban + political + corruption, data = data, vcov = "hetero")
```

```
model3
```

```
## OLS estimation, Dep. Var.: log_fdi
## Observations: 155
## Standard-errors: Heteroskedasticity-robust
##           Estimate Std. Error   t value   Pr(>|t|)
## (Intercept)  14.146653   0.931024 15.19472 < 2.2e-16 ***
## eodb         0.077463   0.013191  5.87256 2.7191e-08 ***
## gdp          0.000051   0.000010  5.04552 1.3074e-06 ***
## unemployment -0.063627   0.022248 -2.85997 4.8496e-03 **
## urban        0.027096   0.006860  3.94987 1.2061e-04 ***
## political    -0.575449   0.198632 -2.89706 4.3390e-03 **
## corruption   -0.456197   0.265204 -1.72017 8.7491e-02 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 1.4501   Adj. R2: 0.540735
```

- What we are doing: Using the natural logarithm of FDI to address skewness in the dependent variable
- Components: Dependent variable: log_fdi, Independent variable: eodb + controls
- Justification: Log transformations stabilize variance and align with similar studies in the literature
- Purpose: To ensure model assumptions are met and improve interpretability of results

```
# Log-Log Model
model4 <- feols(log_fdi ~ log_eodb + gdp + unemployment +
                urban + political + corruption, data = data, vcov = "hetero")
```

```
model4
```

```
## OLS estimation, Dep. Var.: log_fdi
## Observations: 155
## Standard-errors: Heteroskedasticity-robust
##           Estimate Std. Error   t value   Pr(>|t|)
## (Intercept)  2.170085   3.816522  0.568603 5.7049e-01
## log_eodb     4.070701   0.926645  4.392946 2.1193e-05 ***
## gdp          0.000054   0.000010  5.124173 9.1893e-07 ***
## unemployment -0.063750   0.022817 -2.793909 5.8971e-03 **
## urban        0.028214   0.006851  4.118298 6.3232e-05 ***
## political    -0.620405   0.203434 -3.049654 2.7152e-03 **
## corruption   -0.394485   0.267174 -1.476509 1.4193e-01
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 1.46915   Adj. R2: 0.528591
```

- What we are doing: Combining the log-transformation of FDI with control variables to refine the analysis
- Components: Dependent variable: `log_fdi`, Independent variables: `log_eodb` and controls
- Justification: Ensures a comprehensive analysis of EoDB effects while addressing heteroskedasticity and outliers in FDI
- Purpose: To evaluate the robustness of EoDB's impact in a log-transformed context

```
# Interaction Model 1
model5 <- feols(log_fdi ~ eodb * political, data = data, vcov = "hetero")
```

```
model5
```

```
## OLS estimation, Dep. Var.: log_fdi
## Observations: 155
## Standard-errors: Heteroskedasticity-robust
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   13.004850   0.910582 14.28192 < 2.2e-16 ***
## eodb           0.122117   0.013464  9.06979 5.9384e-16 ***
## political     -2.160395   0.716989 -3.01315 3.0329e-03 **
## eodb:political 0.029802   0.010785  2.76328 6.4345e-03 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 1.7182 Adj. R2: 0.368026
```

- What we are doing: Introducing an interaction term between EoDB and political
- Components: Dependent variable: `log_fdi`, Independent variables: `eodb`, `political`, and their interaction
- Justification: Examines if the effect of EoDB on FDI changes under different levels of political stability
- Purpose: To explore conditional relationships influencing FDI inflows

```
# Interaction Model 2
model6 <- feols(log_fdi ~ eodb * corruption, data = data, vcov = "hetero")
```

```
model6
```

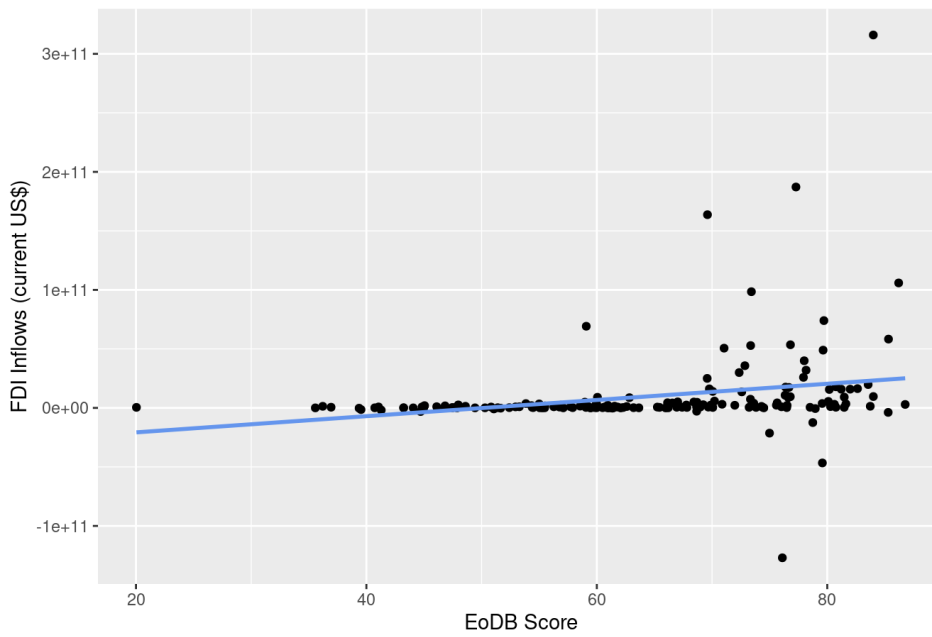
```
## OLS estimation, Dep. Var.: log_fdi
## Observations: 155
## Standard-errors: Heteroskedasticity-robust
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   13.229469   1.180044 11.21100 < 2.2e-16 ***
## eodb           0.116828   0.017071  6.84361 1.8103e-10 ***
## corruption     -2.736972   0.829530 -3.29943 1.2086e-03 **
## eodb:corruption 0.040145   0.010599  3.78750 2.1918e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 1.70283 Adj. R2: 0.379283
```

- What we are doing: Introducing an interaction term between EoDB and corruption
- Components: Dependent variable: `log_fdi`, Independent variables: `eodb`, `corruption`, and their interaction
- Justification: Corruption may moderate the relationship between EoDB and FDI, as countries with better governance might benefit more
- Purpose: To explore governance as a key contextual factor in FDI attraction

Graphs after Regression Analysis

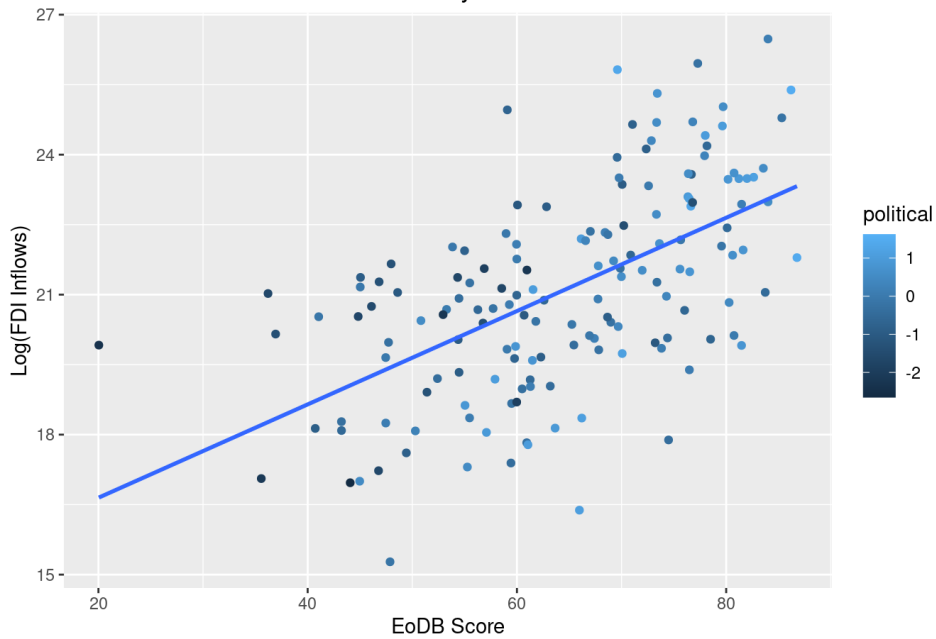
```
# Visualization: Scatterplot with regression line (Simplified Model)
ggplot(data, aes(x = eodb, y = fdi)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "cornflowerblue") +
  labs(title = "FDI Inflows vs Ease of Doing Business", x = "EoDB Score", y = "FDI Inflows (current US$)")
```

FDI Inflows vs Ease of Doing Business

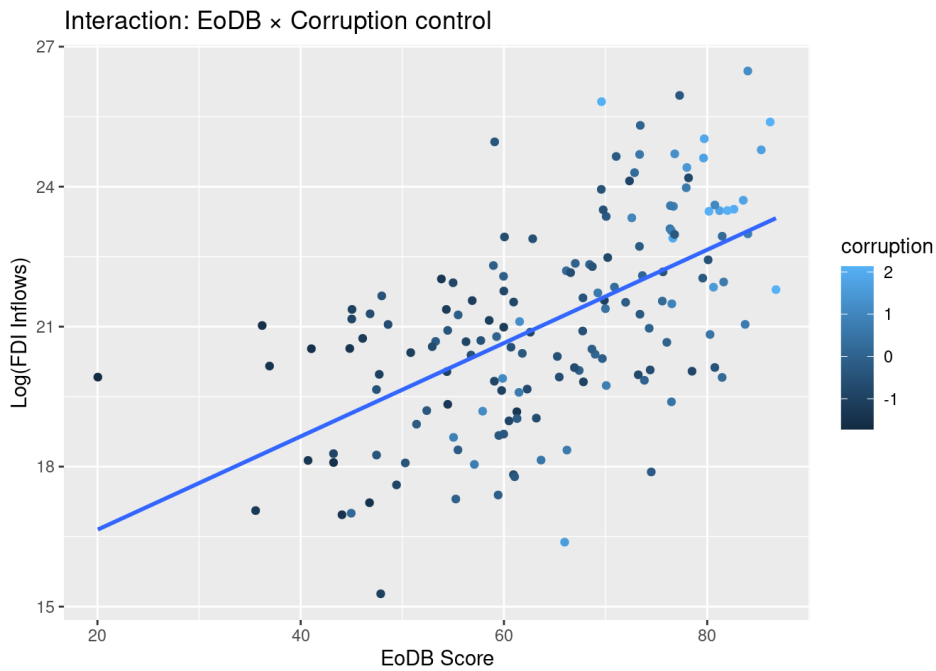


```
# Visualize interaction effects: EoDB × Political Stability
ggplot(data, aes(x = eodb, y = log_fdi, color = political)) +
  geom_point() +
  geom_smooth(method = "lm", aes(group = 1), se = FALSE) +
  labs(title = "Interaction: EoDB × Political Stability", x = "EoDB Score", y = "Log(FDI Inflows)")
```

Interaction: EoDB × Political Stability



```
# Visualize interaction effects: EoDB × Corruption control
ggplot(data, aes(x = eodb, y = log_fdi, color = corruption)) +
  geom_point() +
  geom_smooth(method = "lm", aes(group = 1), se = FALSE) +
  labs(title = "Interaction: EoDB × Corruption control", x = "EoDB Score", y = "Log(FDI Inflows)")
```



- Graphs display regression results, such as the impact of EoDB on FDI and interaction effects (e.g., EoDB × Political Stability).
- Additional plots highlight differences in FDI based on development status.

Creating Summary Table for all regressions

A summary of all regression models is provided using the `etable` function, presenting coefficients, significance levels, and model diagnostics.

```
# Create summary table
models <- list(
  "Simplified Model" = model1,
  "Baseline Model" = model2,
  "Log(FDI) ~ EoDB" = model3,
  "Log(FDI) ~ EoDB + Controls" = model4,
  "Log(FDI) ~ EoDB × Political" = model5,
  "Log(FDI) ~ EoDB × Corruption" = model6)
```

```
# Detailed Regression Results with etable
etable(models, title = "Detailed Regression Results", tex = FALSE)
```

```

##                               Simplified Model
## Dependent Var.:                fdi
##
## Constant                      -3.45e+10* (1.38e+10)
## eodb                          686,404,628.5** (250,786,970.4)
## gdp
## unemployment
## urban
## political
## corruption
## log_eodb
## eodb x political
## eodb x corruption
##
## -----
## S.E. type                      Heteroskedasticity-robust
## Observations                    171
## R2                             0.06190
## Adj. R2                       0.05635
##
##                               Baseline Model      Log(FDI) ~ EoDB
## Dependent Var.:                fdi                log_fdi
##
## Constant                      -3.61e+10* (1.71e+10)      14.15*** (0.9310)
## eodb                          536,030,990.0. (280,719,135.6) 0.0775*** (0.0132)
## gdp                          806,195.0 (539,721.2) 5.08e-5*** (1.01e-5)
## unemployment                 -4,804,530.3 (271,348,672.2) -0.0636** (0.0222)
## urban                       -25,345,998.4 (138,613,167.8) 0.0271*** (0.0069)
## political                   --2,147,483,648.1 (-2,147,483,648.1) -0.5754** (0.1986)
## corruption                   --2,147,483,648.9 (-2,147,483,648.6) -0.4562. (0.2652)
## log_eodb
## eodb x political
## eodb x corruption
##
## -----
## S.E. type                      Heteroskedasticity-robust Heteroskedasti.-rob.
## Observations                    171                      155
## R2                             0.14033                    0.55863
## Adj. R2                       0.10888                    0.54074
##
##                               Log(FDI) ~ EoDB + .. Log(FDI) ~ EoDB .. Log(FDI) ~ EoDB ...1
## Dependent Var.:                log_fdi                log_fdi                log_fdi
##
## Constant                      2.170 (3.817)      13.00*** (0.9106)      13.23*** (1.180)
## eodb                          0.1221*** (0.0135)      0.1168*** (0.0171)
## gdp                          5.38e-5*** (1.05e-5)
## unemployment                 -0.0638** (0.0228)
## urban                       0.0282*** (0.0069)
## political                   -0.6204** (0.2034)      -2.160** (0.7170)
## corruption                   -0.3945 (0.2672)                -2.737** (0.8295)
## log_eodb                     4.071*** (0.9266)
## eodb x political              0.0298** (0.0108)
## eodb x corruption              0.0401*** (0.0106)
##
## -----
## S.E. type                      Heteroskedasti.-rob. Heteroskedas.-rob. Heteroskedas.-rob.
## Observations                    155                      155                      155
## R2                             0.54696                    0.38034                    0.39138
## Adj. R2                       0.52859                    0.36803                    0.37928
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

• What we are doing:

Creating a summary table to consolidate the results of all regression models (model1 to model8) for easier comparison and interpretation.

• Components:

Models: Each model represents a specific variation of the regression analysis (e.g., simplified model, baseline model, log-transformed models, and interaction models).